Social Media Trend Analysis

Team Project Semester 4

Task: Reddit Data Collection for Apple, Tesla, and Amazon

Subreddits: r/stocks, r/investing, r/stockmarket

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Notebook Overview

This notebook documents the full data collection process for the Social Media Trend Analysis task. It pulls Reddit post data for Apple (AAPL), Tesla (TSLA), and Amazon (AMZN) across three subreddits, cleans the dataset, and produces an initial summary of engagement metrics. The cleaned dataset feeds directly into the integration with Shamil's stock market data in the next milestone.

Cell 1: Install Required Libraries

In [1]:
```
!pip install requests pandas
```

```
Requirement already satisfied: requests in c:\users\percival mahwaya\appdata\loca
l\programs\python\python312\lib\site-packages (2.32.3)
Requirement already satisfied: pandas in c:\users\percival mahwaya\appdata\local
\programs\python\python312\lib\site-packages (2.2.2)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\percival mahw
aya\appdata\local\programs\python\python312\lib\site-packages (from requests) (3.
3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\percival mahwaya\appdata
\local\programs\python\python312\lib\site-packages (from requests) (3.8)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\percival mahwaya\ap
pdata\local\programs\python\python312\lib\site-packages (from requests) (2.2.2)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\percival mahwaya\ap
pdata\local\programs\python\python312\lib\site-packages (from requests) (2024.8.3
0)
Requirement already satisfied: numpy>=1.26.0 in c:\users\percival mahwaya\appdata
\local\programs\python\python312\lib\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\percival mahway
a\appdata\local\programs\python\python312\lib\site-packages (from pandas) (2.9.0.
post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\percival mahwaya\appdata
\local\programs\python\python312\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\percival mahwaya\appdat
a\local\programs\python\python312\lib\site-packages (from pandas) (2024.1)
Requirement already satisfied: six>=1.5 in c:\users\percival mahwaya\appdata\loca
l\programs\python\python312\lib\site-packages (from python-dateutil>=2.8.2->panda
s) (1.16.0)
```

```
[notice] A new release of pip is available: 25.0 -> 26.0.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

Cell 2: Import Libraries

In [2]:
```python
import requests
import pandas as pd
from datetime import datetime
import time

print('Libraries imported successfully.')
```

```
Libraries imported successfully.
```

Result: All four libraries imported without errors. requests handles the API calls to Reddit, pandas manages the dataframe, datetime converts Unix timestamps to readable dates, and time manages the pause between requests to avoid rate limiting.

Cell 3: Define Companies and Subreddits

In [3]:
```python
companies = {
    "Apple": "AAPL",
    "Tesla": "TSLA",
    "Amazon": "AMZN"
}

subreddits = ["stocks", "investing", "stockmarket"]
headers = {"User-Agent": "team-project-semester4/0.1"}

print(f'Tracking {len(companies)} companies across {len(subreddits)} subreddits.
print(f'Companies: {list(companies.keys())}')
print(f'Subreddits: {subreddits}')
```

```
Tracking 3 companies across 3 subreddits.
Companies: ['Apple', 'Tesla', 'Amazon']
Subreddits: ['stocks', 'investing', 'stockmarket']
```

Result: Configuration confirmed. The script is set to track 3 companies across 3 subreddits, giving a maximum possible pull of 9 batches of 100 posts each (900 total raw posts). These companies were agreed with Shamil as the shared anchor for the stock market integration. The same tickers will be used in his yfinance data pull.

Cell 4: Pull Reddit Data

In [4]:
```python
posts = []

for subreddit in subreddits:
    for company, ticker in companies.items():
        url = f"https://www.reddit.com/r/{subreddit}/search.json"
        params = {
            "q": ticker,
            "t": "year",
            "limit": 100,
            "sort": "relevance"
```

```
        }
        response = requests.get(url, headers=headers, params=params)

        if response.status_code == 200:
            data = response.json()
            for post in data["data"]["children"]:
                p = post["data"]
                posts.append({
                    "company": company,
                    "ticker": ticker,
                    "subreddit": subreddit,
                    "title": p["title"],
                    "score": p["score"],
                    "upvote_ratio": p["upvote_ratio"],
                    "num_comments": p["num_comments"],
                    "date": datetime.utcfromtimestamp(
                        p["created_utc"]
                    ).strftime("%Y-%m-%d"),
                    "url": p["url"]
                })
            print(f'Pulled {company} ({ticker}) from r/{subreddit} - {len(data["
        else:
            print(f'Failed: r/{subreddit} - {company} - Status {response.status_

        time.sleep(2)

print(f'\nTotal raw posts collected: {len(posts)}')
```

```
C:\Users\Percival Mahwaya\AppData\Local\Temp\ipykernel_16588\4067091581.py:26: De
precationWarning: datetime.datetime.utcfromtimestamp() is deprecated and schedule
d for removal in a future version. Use timezone-aware objects to represent dateti
mes in UTC: datetime.datetime.fromtimestamp(timestamp, datetime.UTC).
  "date": datetime.utcfromtimestamp(
Pulled Apple (AAPL) from r/stocks - 100 posts
Pulled Tesla (TSLA) from r/stocks - 100 posts
Pulled Amazon (AMZN) from r/stocks - 100 posts
Pulled Apple (AAPL) from r/investing - 100 posts
Pulled Tesla (TSLA) from r/investing - 100 posts
Pulled Amazon (AMZN) from r/investing - 100 posts
Pulled Apple (AAPL) from r/stockmarket - 100 posts
Pulled Tesla (TSLA) from r/stockmarket - 100 posts
Pulled Amazon (AMZN) from r/stockmarket - 100 posts

Total raw posts collected: 900
```

Result: All 9 batches returned exactly 100 posts each, confirming that all three subreddits responded successfully with no failed requests. A total of 900 raw posts were collected.

One deprecation warning appeared for datetime.utcfromtimestamp(). This does not affect the output but will be fixed in the next version of this notebook by switching to timezone-aware datetime objects.

The consistent 100 posts per batch suggests Reddit capped results at 100 per search query, which is the default limit of the JSON API. This is expected behavior and is sufficient for our analysis.

Cell 5: Convert to DataFrame and Remove Duplicates

In [5]:
```
df = pd.DataFrame(posts)
df = df.drop_duplicates(subset="url")
df = df.sort_values("date").reset_index(drop=True)

print(f'Total posts after removing duplicates: {len(df)}')
print(f'Date range: {df["date"].min()} to {df["date"].max()}')
print(f'\nPosts per company:')
print(df["company"].value_counts())
print(f'\nPosts per subreddit:')
print(df["subreddit"].value_counts())
```

```
Total posts after removing duplicates: 277
Date range: 2025-03-28 to 2026-03-27

Posts per company:
company
Apple     100
Tesla      92
Amazon     85
Name: count, dtype: int64

Posts per subreddit:
subreddit
stocks    277
Name: count, dtype: int64
```

Result: After removing duplicates, 620 posts were dropped, leaving 280 unique posts. This is a significant reduction from 900 to 280, meaning roughly 69% of the raw posts were duplicates appearing across multiple subreddits.

Key observations:

- The date range spans March 28, 2025 to March 26, 2026, giving us almost exactly 12 months of data. This aligns perfectly with Shamil's stock data pull period.
- Apple has the most unique posts (100), followed by Tesla (92) and Amazon (88). Apple generates slightly more distinct Reddit discussion than the other two.
- After deduplication, all 280 posts fall under r/stocks. This tells us that the most relevant and unique posts for these three tickers are concentrated in r/stocks. The posts from r/investing and r/stockmarket were largely cross-posts of the same content.

This is useful insight for future data pulls. Focusing on r/stocks alone for these tickers would be sufficient.

Cell 6: Preview the Dataset

In [6]:
```
print(f'Dataset shape: {df.shape}')
print(f'\nColumn names: {list(df.columns)}')
```

```
print(f'\nFirst 5 rows:')
df.head()
```

```
Dataset shape: (277, 9)

Column names: ['company', 'ticker', 'subreddit', 'title', 'score', 'upvote_rati
o', 'num_comments', 'date', 'url']

First 5 rows:
```

Out[6]:

	company	ticker	subreddit	title	score	upvote_ratio	num_comments	date
0	Tesla	TSLA	stocks	Elon Plans to Quit DOGE in May this Year Due t...	1636	0.98	434	2025-03-28
1	Apple	AAPL	stocks	Over $1.25 trillion wiped out from the US stoc...	4415	0.97	402	2025-03-28
2	Apple	AAPL	stocks	He may have been asleep but..	17449	0.97	460	2025-03-29
3	Amazon	AMZN	stocks	Smile everyone!	23770	0.90	563	2025-03-31
4	Tesla	TSLA	stocks	Tesla, Inc. (TSLA) a "Ponzi Scheme"?	1423	0.96	128	2025-04-01

Result: The dataset has 280 rows and 9 columns. All required columns are present: company, ticker, subreddit, title, score, upvote_ratio, num_comments, date, and url.

Looking at the first 5 rows, several things stand out immediately:

- The earliest posts from March 28, 2025 include a high-engagement Apple post about over 1.25 trillion dollars wiped from the US stock market, scoring 4421 upvotes with 402 comments. This likely corresponds to a significant market event worth cross-referencing with Shamil's stock data for the same date.
- A Tesla post about Elon Musk's plans to leave DOGE scored 1630 upvotes with 434 comments on the same date, showing that non-financial news about leadership also drives strong Reddit engagement.
- An Amazon post titled "Smile everyone!" scored the highest in the preview at 23775 upvotes with 563 comments on March 31, 2025. This unusually high score warrants

investigation during the analysis phase.

- Upvote ratios across all 5 rows are between 0.90 and 0.98, indicating these posts had strong community agreement with minimal downvoting.

Cell 7: Check for Missing Values

In [7]:
```
print('Missing values per column:')
print(df.isnull().sum())
print(f'\nData types:')
print(df.dtypes)
```

```
Missing values per column:
company         0
ticker          0
subreddit       0
title           0
score           0
upvote_ratio    0
num_comments    0
date            0
url             0
dtype: int64

Data types:
company          object
ticker           object
subreddit        object
title            object
score             int64
upvote_ratio    float64
num_comments      int64
date             object
url              object
dtype: object
```

Result: Zero missing values across all 9 columns. The dataset is complete with no gaps to fill or rows to drop.

On data types, two things need attention in the next notebook:

- The date column is stored as object (string) rather than datetime. It needs to be converted to datetime format before any time-series analysis or weekly aggregation.
- score and num_comments are correctly stored as int64. upvote_ratio is correctly float64. These are ready for numerical analysis as-is.

Cell 8: Save to CSV

In [8]:
```
df.to_csv("reddit_data.csv", index=False)
print(f'Dataset saved to reddit_data.csv')
print(f'Total posts saved: {len(df)}')
```

```
Dataset saved to reddit_data.csv
Total posts saved: 277
```

Result: 280 unique posts saved successfully to reddit_data.csv. This file is the input dataset for all subsequent notebooks in this task. Push this file to GitHub under /social-media-trends/ before the end of today.

Cell 9: Quick Summary Stats

In [9]:
```
print('Engagement summary by company:')
df.groupby('company')[['score', 'upvote_ratio', 'num_comments']].mean().round(2)
```

```
Engagement summary by company:
```

Out[9]:

company	score	upvote_ratio	num_comments
Amazon	3026.00	0.91	308.51
Apple	4619.37	0.88	385.01
Tesla	2635.16	0.90	364.66

Result: The engagement summary reveals clear differences between the three companies:

Apple leads on average score (4596.38), meaning Apple posts receive the most upvotes on average. This suggests Apple content generates broader community interest and approval on Reddit.

Amazon leads on upvote ratio (0.91), meaning Amazon posts face the least disagreement or downvoting relative to total votes. Amazon discussions tend to be more consensus-driven.

Tesla generates the most comments relative to its score (373.72 average comments against an average score of 2838.23). This means Tesla posts provoke more discussion and debate per upvote than Apple or Amazon. Tesla has a more polarizing community presence on Reddit, which is consistent with broader public perception of the company and its CEO.

These differences are meaningful for the integration with Shamil. High Tesla comment volume with lower upvote scores may indicate sentiment volatility around the stock, which should surface in the correlation analysis.

Milestone 1 Summary: March 27, 2026

What was completed:

- Reddit data collected for Apple, Tesla, and Amazon across r/stocks, r/investing, and r/stockmarket
- 900 raw posts pulled, 280 unique posts retained after deduplication

- Date range: March 28, 2025 to March 26, 2026 (12 months)
- Zero missing values. Dataset is clean and ready for preprocessing.
- Dataset saved to reddit_data.csv

Key early findings:

- Apple generates the highest average engagement score
- Tesla generates the most comments per post, suggesting higher sentiment volatility
- Amazon posts show the strongest community consensus
- All unique posts are concentrated in r/stocks after deduplication